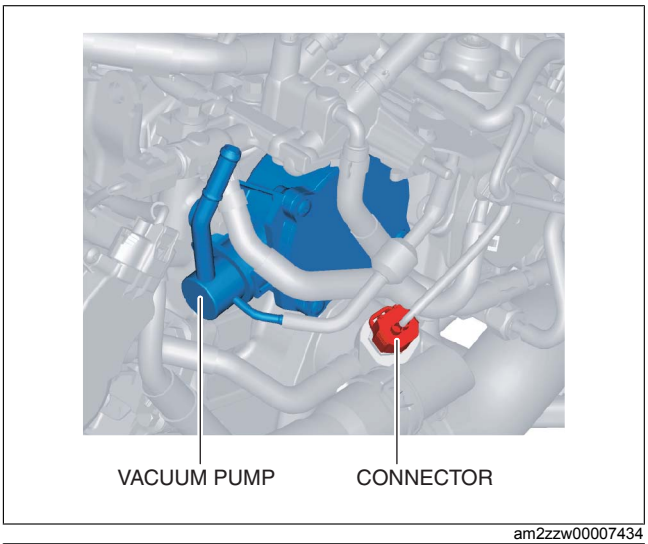
VACUUM PUMP REMOVAL/INSTALLATION [SKYACTIV-D 1.5]

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Warning

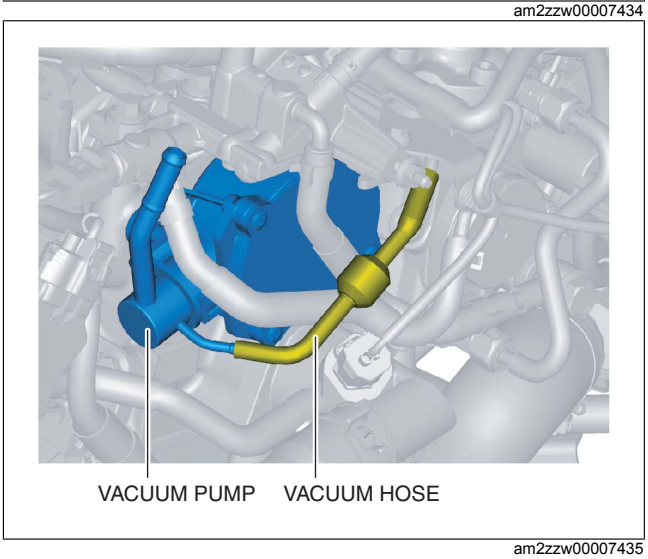
- **A hot engine can cause severe burns. Turn off the engine and wait until it is cool before removing the exhaust system.**

1. Remove the engine cover. (See ENGINE COVER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
2. Remove the battery. (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
3. Pinch open the clamp using pliers and disconnect the vacuum hose from the vacuum pump. (See VACUUM HOSE REMOVAL/INSTALLATION [L.H.D.].) (See VACUUM HOSE REMOVAL/INSTALLATION [R.H.D.].)
4. Remove the following parts as a single unit. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
   - Fresh-air duct
   - Air hose
   - Air cleaner cover
   - Air cleaner element
   - Air cleaner case
   - Air inlet pipe
5. Remove the vacuum pipe installation bolts. (See FUEL INJECTOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
6. Disconnect the engine coolant temperature (ECT) sensor connector.



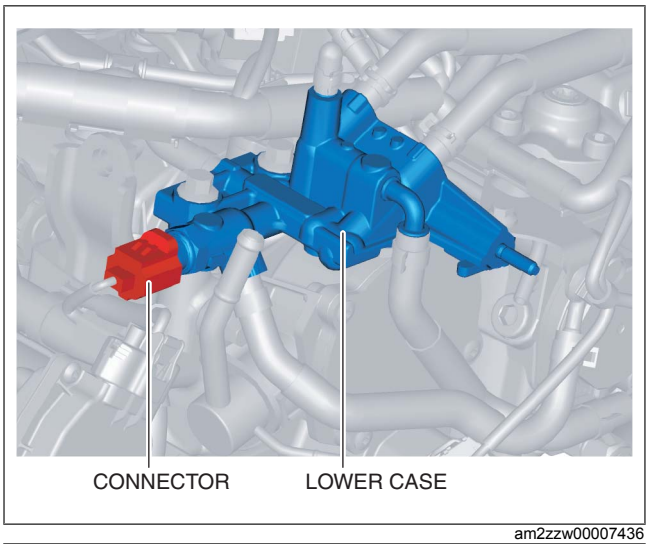
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7. Remove the vacuum hose.



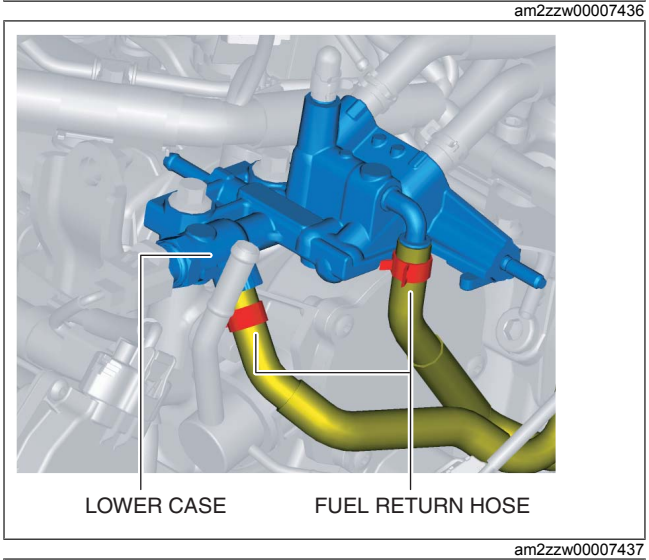
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8. Disconnect the fuel temperature sensor connector.
9. Perform the “BEFORE SERVICE PRECAUTION”. (See BEFORE SERVICE PRECAUTION [SKYACTIV-D 1.5].)



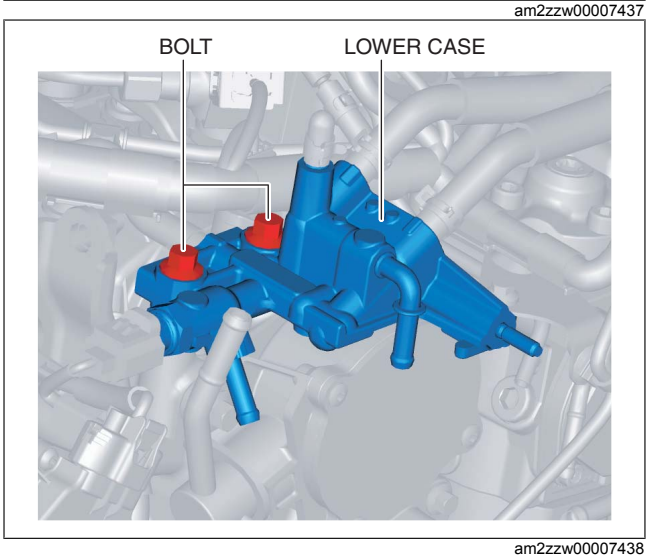
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10. Disconnect the fuel return hoses from the lower case. (See LOWER CASE REMOVAL/ INSTALLATION [SKYACTIV-D 1.5].)

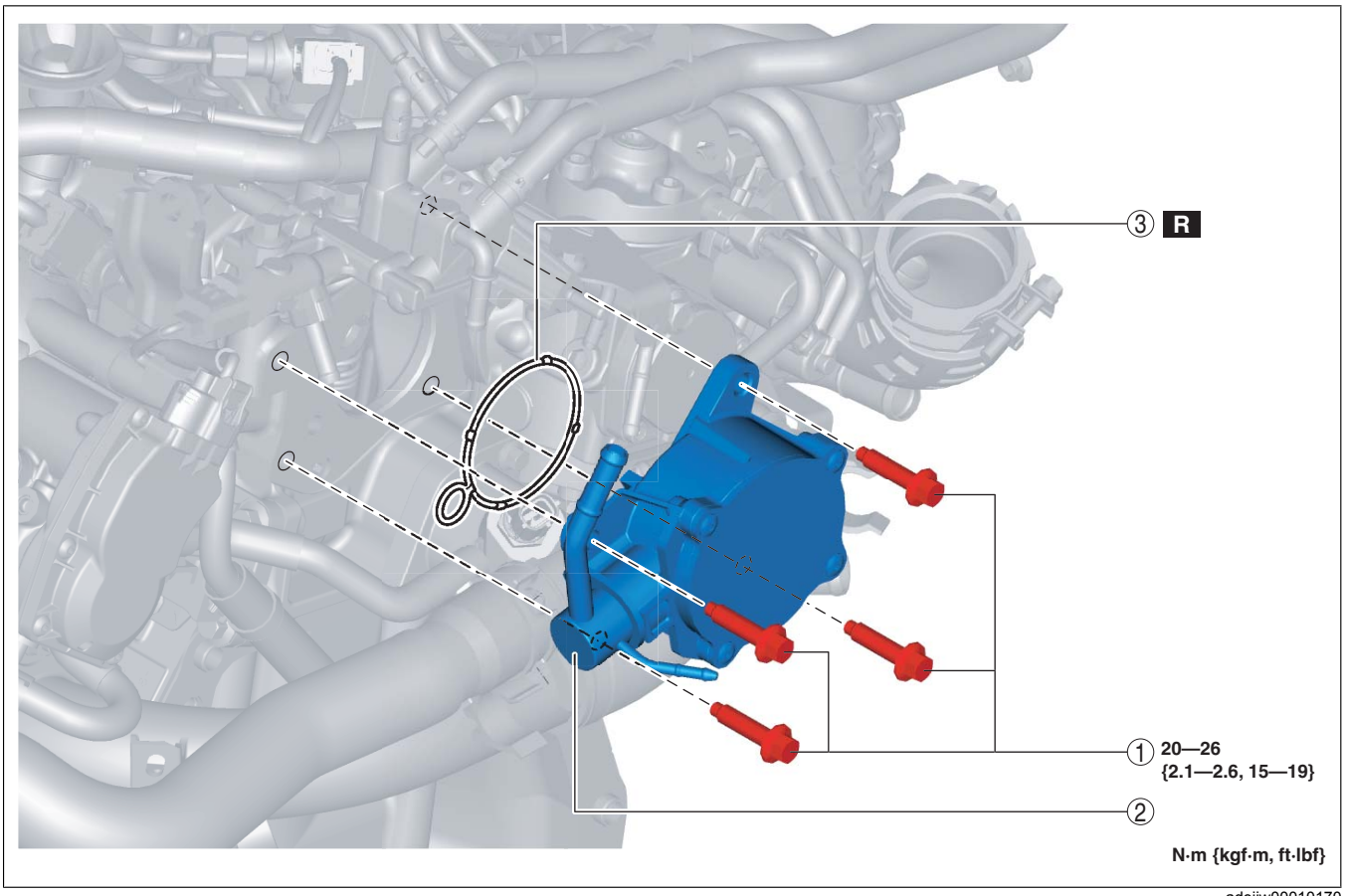


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11. Remove the lower case installation bolts. (See LOWER CASE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
12. Remove in the order shown in the table.
13. Install in the reverse order of removal.
14. Perform the “AFTER SERVICE PRECAUTION”. (See AFTER SERVICE PRECAUTION [SKYACTIV-D 1.5].)



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1	Bolt
2	Vacuum pump (See Vacuum Pump Removal Note.) (See Vacuum Pump and O-ring Installation Note.)
3	O-ring (See Vacuum Pump and O-ring Installation Note.)

Vacuum Pump Removal Note

1. When removing the vacuum pump, place a rag under the installation surface of the vacuum pump so that engine oil does not get on other parts.
2. Remove the vacuum pump.

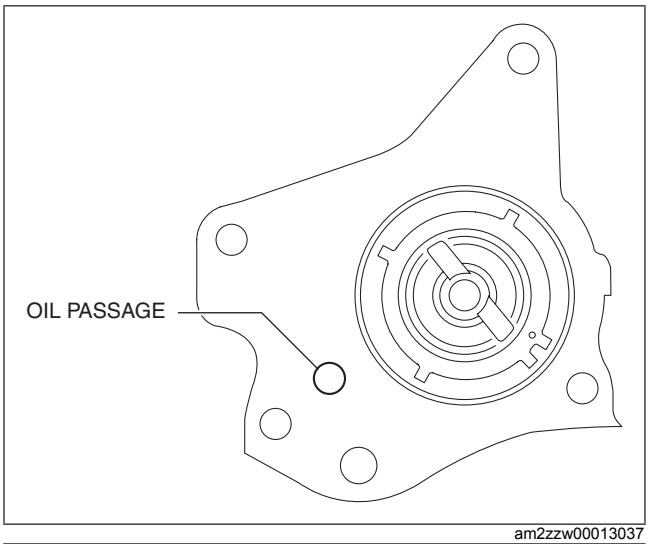
Vacuum Pump and O-ring Installation Note

1. Rotate the vacuum pump by hand to drain the remainder of the engine oil in the vacuum pump.
2. Degrease the O-ring installation groove and installation surface.
3. Install the new O-ring to the vacuum pump.

Caution

- **When cleaning away the remaining silicone sealant on the cylinder head, insert a cloth into the oil passage and be careful not to allow silicone sealant to penetrate the oil passage. Otherwise, silicone sealant may penetrate the oil passage.**
- **Use a cloth when blocking the oil passage. If a shop paper towel is used, pieces of the oil soaked towel may break off and enter the oil passage.**

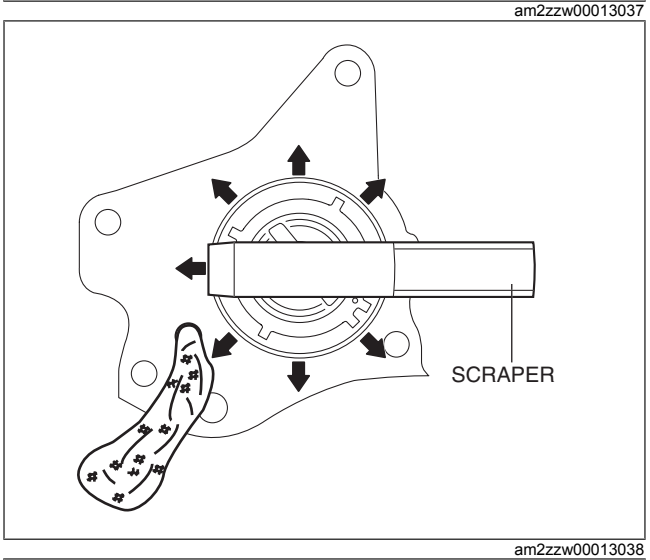
4. Insert a cloth into the oil passage shown in the figure.



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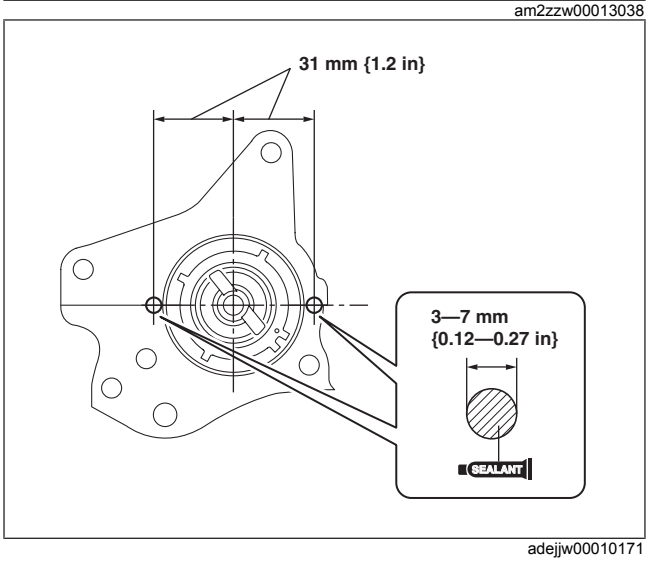
5. Scrape the cylinder head from the inside to the outside using a scraper and clean away the remaining silicone sealant on the cylinder head so that it does not get in the cylinder head.
   - If silicone sealant gets in the cylinder head, blow it out using compressed air.
6. Clean the cylinder head.

Caution
- **Be careful not to apply a large quantity of silicone sealant. Otherwise, the silicone sealant may overlap the O-ring of the vacuum pump and block the oil passage of the cylinder head.**
- **Install the vacuum pump before silicone sealant dries.**



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7. Apply silicone sealant (TB1217D or equivalent) to the positions shown in the figure.
8. Install the vacuum pump.



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